

Trajectory Optimization for Drone Logistics Delivery via Attention-Based Pointer Network

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Abstract—Drone logistics delivery is a potential booster to redefine the logistics system efficiency, which has been a new special hot research field. Among that, how to optimize drone trajectory data and find optimal delivery path is a crucial problem. However, most existing studies fail to derive the feasible trajectory data due to underestimating drone energy consumption and delivery multi-variant constraints for air transportation. In this paper, we propose a novel self-driven learning procedure, named attention-based pointer network (A-Ptr-Net) model, which can solve drone delivery trajectory optimization problem. The generated A-Ptr-Net model coupled with attention mechanism is effective on adapting to the new drone trajectory data automatically, regardless of explicit distance matrix. We go into developing the convex function constraints related to drone non-linear energy consumption, customer demand and service time windows, which is applied on A-Ptr-Net model for optimizing the drone logistics delivery. Numerical experiments indicate that the proposed method performs significantly better than classical heuristics for drone trajectory analysis and optimization.

Index Terms—Drone logistics delivery, trajectory optimization, pointer network model, attention mechanism.

I. INTRODUCTION

RECENTLY, drone delivery has attracted great attention, which is known as a potential booster to redefine logistics system efficiency. Amazon put forward Amazon Prime Air to promote the quality of providing delivery service for customers with drones. Subsequently, many well-known organizations such as Google, Walmart and Federal Aviation Administration (FAA) also started delivering parcels by testing drones [1], [2]. In Fig. 1, it is a typical overview of the drone

logistics delivery. Compared with traditional truck delivery, the advantages and characters of drone delivery are obvious: 1) Drones can reduce delivery time, and customers can get service earlier than truck delivery; 2) Drones are less expensive to be run than trucks, so they minimize total operation costs; 3) Drones use less labor to perform delivery tasks, and the labor costs are saved; 4) Drone delivery is more flexible in emergency situations, such as earthquakes, forest fires and flood disaster. Drones get rid of the restriction of transportation network for vehicle routing. However, challenges of drone delivery limit its development to some extent. Therefore, drone trajectory data analysis is worth further studying through scientific means.

Significant effort has gone into analyzing and optimizing drone trajectory data problem. Many scholars have shown deep interests in using exact and heuristic algorithms to overcome the challenges. The flying sidekick traveling salesman problem (FSTSP) turned on an important beginning for researching the drone trajectory optimization [3]. On this basis, ‘cluster first-route second algorithm’, ‘interactive programming’ and ‘saving or insertion model’ also raised some feasible vehicle routing projects, in the event of limited flight range and small items of drones [4], [5]. In general, the mathematical exact algorithm gets more accurate calculation results, but the experimental samples are very small, which is a big problem. Heuristic algorithm is faced with the difficulty of not proving the optimality of the results, although dealing with large-scale experiments. It can be seen that how to find a suitable method to optimize drone logistics delivery is still an obstacle to its development.

Deep reinforcement learning has been a compelling choice to solve complex tasks with trajectory data. Especially for combinatorial optimization problem, deep reinforcement learning has an incomparable superiority than ‘exact or heuristic algorithm’ [6]. Most combinatorial optimization problems are related to decision-making order, which can be regarded as sequential decision-making problem. For example, for traveling salesman problem (TSP), it is to decide the order to visit each city. Deep neural network can just complete the mapping from one sequence to another, so it is a feasible way to directly solve drone delivery routing optimization problem by defining network state and reward [7]. Deep reinforcement learning does not require any trajectory data to be given in advance, but obtains the training information and updates the parameters by receiving the feedback from

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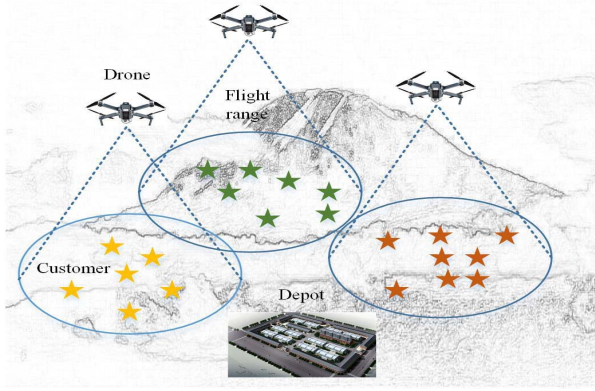


Fig. 1. Overview of the drone logistics delivery.

the iteration, so as to deal with the trajectory data sequential decision-making.

However, most existing studies on drone logistics delivery are defective by the randomness of trajectory data. These problems can be summarized as follows: 1) Classical heuristic algorithm requires an definite distance matrix when calculating the optimal delivery trajectory. If the data is updated, the results cannot adapt to the solution automatically. In addition, the model may lose the global optimality, in case of dealing with trajectory data; 2) To simplify the calculation of drone energy consumption, existing researches make the linear approximation for nonlinear energy function based on its payload and battery weight. This approach may fail to find feasible delivery routes, underestimating the drone flight environment; 3) As drone trajectory data optimization is NP-hard problem, few large instances are tested. Multiple depots and customer nodes take more time to search for information, and the model structure is more uncontrollable. In order to overcome these shortcomings, some efforts must be made.

In this paper, the A-Ptr-Net model coupled with attention mechanism is adopted to optimize drone trajectory data, in order to find optimal delivery path. We take drone energy consumption, customer demand and service time windows into account, and then an generated network model is proposed. The following three major contributions can be summarized:

1) We propose a novel self-driven learning procedure, named A-Ptr-Net model, which is constructed by a set of integral generated sequences to analyze and optimize drone trajectory data. Unlike heuristic methods, the distance matrix need be recalculated when data is updated. To fill these gaps, the proposed reinforcement learning model can derive the reward calculation based on feed-forward trajectory data unit. It means that the decoder output can adapt to the new trajectory data automatically, regardless of explicit distance matrix. Even though we make no clear the calculation process but can gain cost of the given solution, we can come up with a feasible trajectory using our model.

2) Attention mechanism is added to the decoder context, improving the search performance of trajectory data. It consists

of multi-head attention layer and fully connected feed-forward layer. Attention mechanism allows the encoder to multiple output instead of using a single long vector of the corresponding step, so more information is retained. Furthermore, it helps the decoder to consider hidden state sequence of the whole encoder output, so that the graph embedding can store more information in all hidden state vectors. When decoder context uses these hidden vectors, it can decide which vectors to pay more attention to.

3) Compared with the linear approximation for energy consumption function, this paper considers keeping the non-linear with the drone parameters. Existing studies on drone energy consumption ignore energy consumption differences in takeoff, landing, flight and hovering. We construct a convex function related to drone payload, battery weight and own weight, which can reduce the actual error. In addition, service time windows ensure the drone arrival time and service time by most reasonable.

The rest of this paper is organized as follows. In Section II, we give the related literature review on drone logistics delivery problem and application of deep reinforcement learning in analyzing trajectory data. In Section III, drone logistics delivery based on trajectory optimization are given in formulation. Section IV proposes the A-Ptr-Net architecture for drone trajectory analysis. In Section V, the experimental results are discussed. Section VI concludes this paper.

II. RELATED WORK

In order to further analyze the related studies on drone logistics delivery based on trajectory data, we will discuss it in three subsections. Specially, the drone logistics delivery problem and its variants will be summarized. Exact and heuristic algorithms are discussed for optimizing drone logistics trajectory. Application of deep reinforcement learning in analyzing trajectory data will be also discussed.

A. Drone Logistics Delivery Problem

The research on drone logistics delivery problem appeared for the first time in the literature of traveling salesman problem (TSP). Murray and Chu proposed two new variants of the traditional TSP called the parallel drone scheduling traveling salesman problem (PDSTSP) and the flying sidekick traveling salesman problem (FSTSP). The FSTSP model can meet a series of customer needs under considering the minimum service time [3]. Based on this model, Agatz *et al.* considered the truck collaborates with a drone to analyze TSP-D [8], Huang *et al.* [9], and Freitas and made use of the variable neighborhood search to solve the flying sidekick traveling salesman problem [10], [11]. The vehicle routing problem (VRP) is complicated with the traveling salesman problem, which consists of several related TSP subproblems. The vehicle routing problem with drone is to study how to arrange the vehicles route, so that the drone can serve each customer in turn according to the shortest path [12], [13]. In the delivery system for VRP, many dynamic variants related to drone need to be considered during completing multitrip tasks, compared with the traditional only truck delivery.

Drone logistics delivery belongs to a multi-variant combinatorial optimization problem. Ha *et al.* put forward a multi-objective variant of the flying sidekick traveling salesman problem [14]–[16]. They made use of the integer programming to solve the multi-variant objective optimization, considering the delivery completion time and the total operational cost. Salama also considered two main variants for the flying sidekick traveling salesman problem with drone, and they proposed the integrated optimization approach to choose a cluster delivery locations [17], [18]. In general, minimizing the total operational cost and the completion time perform effective local search and genetic search results [19]–[21]. Multi-variant objective optimization depends on setting reasonable mathematical functions and developing a exact or heuristic VRP modeling.

Different from truck delivery only, multitrip drone delivery faces more variants, such as payload capacity, flight endurance, travel time and battery capacity. To take payload capacity into account, drone may carry small items or a few deliveries per trip, which increase the final completion time of drone delivery [22]. In the cooperative delivery of truck and drone, drone need to return to truck for loading parcels many times. The flight endurance and travel time determine the operation time, so the drone is restricted from leaving or returning to depot. Chase and Murray minimized the overall customer waiting time by launching from the same place, and they presented a heuristic algorithm to solve real-world-delivery instances [23], [24]. In the variant of drone delivery, battery capacity is a whole new issue for multitrip drone delivery, and this is a hard nut to crack. Limited flight range makes drone be within a certain range of energy supply, and the number of drone service customers is limited [25]. Besides, weather uncertainty adds the travel risk and the cancelations of services.

Energy consumption function of multitrip drone logistics delivery is subject to nonlinearity, and the flight distance is affected by battery capacity, weather conditions and other complex factors. In order to simplify the calculation process, many researchers tend to make a linear approximation for drone flight distance. Dorling *et al.* derived the energy consumption that the drone status includes in hover, flight, takeoff and landing during logistics distribution. Linear approximation of the drone energy consumption is compelling choice [26]–[28]. Some other researchers kept the energy consumption function in nonlinearity, but the instances number is only about 100, which has a big gap with the actual situation [10], [29]. Therefore, how to keep drone enough energy is the key issue to solve the multitrip drone logistics delivery problem.

Service time windows variable is closely related to consumer experience. For TSP and VRP, the service time of each customer is random and needs to meet a certain interval. The upper and lower limits are the latest and earliest acceptable arrival time respectively. Early or late arrival both affects customer satisfaction, because too early and too late may lead to customer dissatisfaction [30]. In the logistics distribution system, service time windows is a kind of condition restriction, which increases the difficulty of solving the model. According

to the time constraint conditions, VRP can be divided into two categories: soft time windows VRP and hard time windows VRP [31]. The soft time windows VRP requires that the access should be reached within the time windows as much as possible, otherwise, some punishment will be given. If the vehicle arrives before the customer's earliest arrival time, the cost of waiting must be borne. If the vehicle arrives after the latest required arrival time, a penalty value will be imposed. The hard time windows VRP requires that the access must arrive within the time windows, otherwise the service will be denied immediately.

B. Exact and Heuristic Methods for Trajectory Optimization

Trajectory data analysis and optimization is known to be NP-hard problem. Exact and heuristic algorithms are common methods to analyze trajectory data [32], [33]. For exact algorithm, 'branch and bound', 'integer programming' and 'dynamic programming' can get the solution by constructing rigorous mathematical model or using data structure search. The branch and bound expands the feasible solution of the problem into branches like a tree, and then it searches the best solution in each branch. Li *et al.* proposed the 'branch and bound' for the synchronized logistics delivery problem with trajectory data, and the numerical results confirmed the effectiveness [34]. 'integer programming' can be done by adding integer variables to the mathematical model and listing the objective equation to limit the trajectory data. 'Integer programming' takes the actual situation as constraints into the model, which makes complex problems easier to understand [35], [36]. Eshragh *et al.* built the dynamic programming approach for the logistics delivery problem based on trajectory data. They decomposed a problem into several small problems and got the shortest distance between the two points in the path through the reverse solution [37], [38]. However, all of above algorithms are only suitable for small batch and limited calculation range.

Heuristic algorithm is another important method, genetic algorithm, simulated annealing algorithm, tabu search and ant colony optimization are widely used to solve logistics delivery problem based on trajectory data from different direction. Although the heuristic algorithm provides a new solution possibility, it cannot guarantee the global optimal solution. For example, Poikonen and Dayarian considered the mothership and drone logistics delivery by second-order cone heuristics program. Under multi customer samples, heuristic algorithm can solve the feasible solution of instances up to 200 customers at an acceptable cost [39], [40]. Genetic algorithm is a classical intelligent optimization algorithm for solving discrete trajectory data optimization problem. It obtains diverse solutions through crossover operation and mutation operation, and it retains the optimal solution of the objective function in the next generation. It selects the next generation of individuals by roulette, so as to continuously optimize the solution [41]. It is thus clear that genetic algorithm fails to tremendous computational complexity, especially in searching optimal solution of multi-objective trajectory data.

C. Application of Deep Reinforcement Learning in Analyzing Trajectory Data

Deep reinforcement learning is the advanced topic in analyzing and optimizing drone trajectory data problem. In recent years, deep reinforcement learning has achieved satisfied results in many target recognition and prediction tasks. Therefore, it is a promising choice to apply deep reinforcement learning to combinatorial optimization problem. Specifically, the essence of trajectory optimization problem is sequential decision-making order, and it is to decide in what order to access customer nodes for VRP [47]–[49]. Deep reinforcement learning is born to be used for sequential decision-making, so the sequential decision-making in trajectory optimization problem can also be solved accurately. The difficulty is how to define state and reward of deep reinforcement learning algorithm.

Pointer network is a neural network designed for the trajectory optimization. Pointer network is derived from the sequence to sequence network structure proposed by Google [43]. Firstly, the network imports the data into the encoder, and then it returns the pointer through the decoder pointing to the position or serial number of the encoder output, as the name pointer network. Bello *et al.* tried to solve the traditional TSP by constructing pointer network, and they use policy gradient and actor-critic to train network and achieve preliminary experimental results [44]. Base on it, Nazari *et al.* and Dai *et al.* extended the pointer network to solve the VRP from TSP. Pointer network is insensitive to the order of the input data, which simplifies the coding process and makes embedded representation [45], [46]. Adding attention mechanism to pointer network is a new way to solve combinatorial optimization problem. In the VRP with truck delivery, attention mechanism can establish a better connection between input and output sequence. Consequently, pointer network has achieved initial and innovative results than exact and heuristic algorithms for VRP.

To sum up the representative studies of drone logistics delivery and application of deep reinforcement learning in analyzing trajectory data, Table I makes a list about objective function, drone energy, drone capacity, time windows, approach and so on.

III. FORMULATION OF DRONE LOGISTICS DELIVERY

In this section, we will make the formulation for drone logistics delivery. The actual meaning of drone logistics delivery problem will be represented by some mathematical symbols. Assumptions help us to construct a reasonable mathematical model about multitrip drone delivery, and then we will develop the decision variables and constraints in details. Finally, objective function will be defined to facilitate the model solution in the next step.

A. Assumptions

We assume that logistics distribution center called depot provides the parcel delivery service for drone. Each drone has the limited payload, and only one trip cannot meet the demands of all customers. At a time, it can only deliver

several parcels, which is a very small quantity relative to the total tasks. Therefore, drone needs to return to the depot to load new parcels with recharging, and depot has prepared enough batteries for drone to provide energy. Each customer demand node exists the fixed service time windows, which requires drone to complete delivery tasks within the specified time. In addition, we assume that the flight speed of drone is constant and it is not affected by weather conditions. The time to replace batteries, load parcels at depot and unload parcels at customer nodes is ignored.

Regarding to drone flight condition and delivery tasks, the specific assumptions are as follows:

1) Each customer demand node is determined. $N_0 = \{1, 2, 3, \dots, n\}$ represents all the customer nodes, and it is the set of logistics demand. All the nodes obey the random distribution.

2) Logistics delivery network can be defined as: $N = \{0, N_0\}$, which is the set of all customer demand nodes and a depot. Node 0 represents the depot, and it is responsible for parcel storage and drone energy supply.

3) r_i ($\forall i \in N_0$) is the number of customer delivery demand. The number of demand is random and cannot be predicted in advance, and it makes sure the rationality of the model application.

4) The service time windows of customer node is set as: $[a_i, b_i]$. a_i represents the earliest possible arrival time of node i , and the latest possible departure time with b_i . The service time windows is a unique time constraint factor.

5) The maximum drone capacity of each trip is expressed as ζ , and it is constrained by $\zeta \geq r_i, \forall i \in N_0$. The demand of each customer is less than the maximum drone capacity, and this ensures that every customer node can be served.

6) t_{ij} is the flight time during multitrip drone delivery, associated with network node i and j .

B. Decision Variables and Constraints

1) *Decision Variables*: The model contains a binary variable v_{ij} and a location variable x_i . Correspondingly, $v_{ij} = 1$ indicates that the drone moves from customer node i to node j , and $v_{ij} = 0$ indicates that drone distribution network is interrupted. The location of customer demand node and depot is represented by $x_i, \forall i \in N_0$ and x_0 respectively.

There are four sets of continuous variables: ϕ_i , e_{ij} , g_{ij} and ε_i . ϕ_i is the drone energy consumption when it arrive at customer node i . e_{ij} is the energy consumption during the flight course (i, j) . g_{ij} is the parcel weight carried by drone fighting in network (i, j) . ε_i indicates the start of service time at node i . Above decision variables make sure that drone completes all the delivery services.

2) *Network Graph Constraints*: In order to ensure that every route is valid through, network graph constraints are expressed as Eq. 1 - Eq. 3.

$$\sum_{j \in N, i \neq j} v_{ij} = 1 \quad \forall i \in N_0 \quad (1)$$

$$\sum_{j \in N, i \neq j} v_{ji} = 1 \quad \forall i \in N_0 \quad (2)$$

TABLE I
SUMMARY OF STUDIES ON VRP WITH DRONE

Authors	Objective function	Drone energy	Drone capacity	Multiple visits	Time windows	Approach	Instances
<i>Murray and Chu</i> [3]	completion time	no	no	no	no	heuristic	up to 10 customers
<i>Agatz et al.</i> [8]	operations costs	no	no	no	no	heuristic	up to 100 customers
<i>Huang et al.</i> [9]	traversal time	yes	no	no	no	heuristic	up to 6 customers
<i>Cheng et al.</i> [10]	travel distance	yes	yes	no	yes	exact	Solomon(1987) and Dorling(2017)
<i>Freitasand and Penna</i> [11]	completion time	no	no	no	no	heuristic	Ponza(2016) and Agatz(2018)
<i>Patchara et al.</i> [42]	delivery costs	no	yes	yes	no	heuristic	up to 150 customers
<i>Wang et al.</i> [12]	completion time	yes	yes	no	no	heuristic	up to 500 customers
<i>Ha et al.</i> [14]	route costs	no	yes	no	no	heuristic	up to 100 customers
<i>Meng et al.</i> [15]	travel distance	no	no	yes	no	heuristic	up to 20 customers
<i>Salama and Srinivas</i> [17]	delivery costs	no	yes	yes	no	heuristic	up to 35 customers
<i>Xu et al.</i> [18]	route cost	no	no	no	no	heuristic	Reinelt(1995)
<i>Bouman et al.</i> [22]	completion time	no	no	no	no	exact	up to 20 customers
<i>Murray and Raj</i> [23]	completion time	no	yes	no	no	heuristic	up to 100 customers
<i>Poikonen and Golden</i> [24]	operations costs	no	no	no	no	heuristic	up to 200 customers
<i>Dorling et al.</i> [26]	operations costs	yes	yes	yes	no	exact	up to 50 customers
<i>Kirschstein</i> [27]	completion time	yes	yes	no	no	heuristic	Bundsserband(2017)
<i>Jeong et al.</i> [29]	completion time	no	no	no	no	heuristic	up to 10 customers
<i>Daniel et al.</i> [36]	travel distance	no	yes	yes	yes	exact	Hernandez(2016) and Catta(2016)
<i>Mehlawat et al.</i> [41]	completion time	no	yes	no	no	heuristic	up to 50 customers
<i>Vinyals et al.</i> [43]	completion time	no	no	no	no	reinforcement learning	up to 500 customers
<i>Bello et al.</i> [44]	travel distance	no	no	yes	yes	reinforcement learning	Kingma(2014)
<i>Nazari et al.</i> [45]	route length	no	no	yes	yes	reinforcement learning	Laporte(1992) and Gendreau(2000)
<i>Dai et al.</i> [46]	completion time	no	no	no	no	reinforcement learning	up to 100 customers

$$\sum_{j \in N, i \neq j} v_{ij} - \sum_{j \in N, i \neq j} v_{ji} = 0 \quad \forall i \in N \quad (3)$$

Eq. 1 and Eq. 2 guarantee that every customer node is visited, except for the depot. Eq.3 can prevent the repeated delivery for a certain node i . The delivery routing map uses the binary variable v_{ij} to structure a integrated drone service network.

3) *Energy Consumption Constraints*: Dorling *et al.* suggest that drone energy consumption $\zeta(g_{ij})$ is nonlinear [26]. This is because the drone energy consumption in hover state is greater than that in flight, take-off or landing. On average, the multirotor helicopter is reduced due to translational lift. Energy consumption is a convex function related to drone payload, battery weight and own weight. We use Eq. 4 to represent the drone energy consumption:

$$\zeta = (W + m + g_{ij})^{3/2} \sqrt{\frac{\chi^3}{2\rho\varrho\sigma}} \quad (4)$$

where W , m and χ are drone frame weight, battery weight and gravity, respectively. ρ and ϱ are the fluid density of wind and the area of spinning blade rotor. σ represents the number of

helicopter rotor. Further, Eq. 4 can be rewritten as Eq. 5, if it exacts $\lambda = \sqrt{\frac{\chi^3}{2\rho\varrho\sigma}}$:

$$\zeta = \lambda (W + m + g_{ij})^{3/2} \quad (5)$$

Since the environment impact on drone energy consumption is not discussed in this study, λ is a fixed constant and it cannot be considered. Therefore, the energy consumption constraints are expressed as follows:

$$\varphi_i + \lambda' (W + m + g_{ij})^{3/2} t_{ij} \leq \varphi_j + H_{ij} (1 - v_{ij}) \quad (6)$$

$$H_{ij} = \lambda' (W + m)^{3/2} t_{ij} + \eta \quad (7)$$

$$r_0 = 0, \quad r_{n+1} \leq \zeta \quad (8)$$

Eq. 6 and Eq. 7 indicate the functional relationship of energy consumption between network node i and its next delivery node. η is the energy quantity of drone carried each time. The constant related to gravity, fluid density of wind, area of spinning blade and number of rotors is expressed as λ' . Eq. 8 ensures that drone energy consumption is 0 at depot, that is, the drone needs to be calculated the energy consumption again after return.

4) *Customer Demand Constraints*: Consumer demand constraint is an important factor for multitrip drone delivery. According to different demand information of nodes, drone plans the optimal vehicle routing. The formula of customer demand constraints are shown as Eq. 9 - Eq. 11:

$$\sum_{j \in N, i \neq j} g_{ji} - \sum_{j \in N, i \neq j} g_{ij} = r_i \quad \forall i \in N_0 \quad (9)$$

$$g_{ij} \leq \zeta v_{ij} \quad \forall (i, j) \in N \times N \quad (10)$$

$$g_{i,n+1} = 0 \quad \forall i \in N_0 \quad (11)$$

Considering the drone limited payload for each trip, the needs of all consumers in the distribution network are met, and there is no repeated or failure delivery. In addition, drone cannot carry parcels from customer demand nodes to depot.

5) *Service Time Windows Constraints*: Service time windows constraints are enforced by

$$\varepsilon_i + t_{ij} - H''_{ij} (1 - v_{ij}) \leq \varepsilon_j \quad \forall (i, j) \in N_0 \quad (12)$$

$$\varepsilon_i + t_{i,n+1} + t_{0j} \leq \varepsilon_j + H''_{ij} \quad \forall (i, j) \in N_0 \quad (13)$$

$$a_i \leq \varepsilon_i \leq b_i \quad \forall i \in N_0 \quad (14)$$

The service time windows constraints of different customer nodes are considered, the above inequality ensures that the drone completes the delivery tasks within the specified time. If it exists $v_{i0} = 0$, the earliest possible arrival time is also zero $a_i = 0$, $a_i > 0$ otherwise. By setting the parameters of drone flight time, service time and flight continuity, the service time windows requirements of each customer node are met.

C. Objective Function

To solve the multitrip drone logistics delivery based on trajectory data, the objective function is to minimize the total route distance of drone completing all the parcel delivery tasks.

$$\begin{aligned} \min d \\ \text{s.t. } (1) - (14) \end{aligned} \quad (15)$$

where d represents the drone total delivery distance. The goal of the model is to pursue the minimum drone flight distance, that is, the optimal logistics delivery path. At the same time, the function needs to meet the constraints constructed above. Eq. 15 shows the objective function whit drone completing the delivery task under constraints.

Above combinatorial optimization belongs to be NP-hard, especially the nonlinear constraints in group (1) – (14) make it more difficult to solve exactly. Solution methods can be divided into exact algorithm and heuristic algorithm. Exact algorithm provides the optimal guarantee solution, but it is unable to process large-scale examples. Heuristic algorithm can only get sub optimal solution because of no accurate theory. Considering the trade-off between optimality and computational cost, the heuristic algorithm can find the suboptimal solution in the acceptable running time of large-scale examples. However, designing heuristic algorithm is a tedious process, especially it is not easy to design heuristic algorithm with good performance.

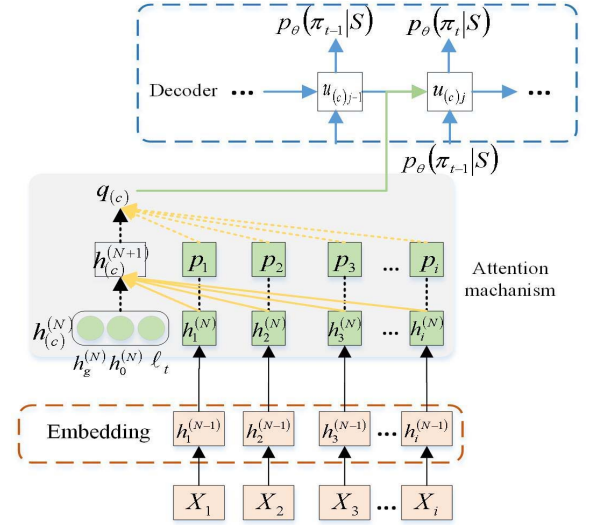


Fig. 2. The proposed attention-based pointer network model structure.

Deep reinforcement learning has achieved remarkable results in trajectory optimization. We can use the deep reinforcement learning method to design a complete end-to-end neural network. It learns the hidden heuristic information without manual intervention automatically, so as to achieve the effect similar to the exact algorithm as far as possible. At the same time, it also overcomes the problem that the exact algorithm cannot calculate the complex function.

IV. ATTENTION-BASED POINTER NETWORK ARCHITECTURE FOR DRONE TRAJECTORY OPTIMIZATION

The essence of drone trajectory optimization is to decide the trajectory node sequence order, and A-Ptr-Net model is an end-to-end neural network model to solve the trajectory sequence decision-making in order. A-Ptr-Net model is a new network architecture inspired by sequence to sequence network, which solves the mapping problem from one trajectory sequence to another. The content of output sequence is exactly the same as that of trajectory input sequence, but the sequence has changed. Therefore, A-Ptr-Net model establishes the fantastic connection between neural network and drone trajectory optimization problem.

A-Ptr-Net model uses one recurrent neural network (RNN) to map the trajectory input to an embedding, and then it uses another RNN to map the embedding into the trajectory output. These two RNNs are called encoding and decoding respectively. In addition, an attention mechanism is added to the model framework, which establishes the relationship between an element of the output sequence and elements of the input sequence. When calculating the output sequence, it will get the weight associated with the element position of the output sequence. The position weight of the input sequence is represented, and the input sequence and the position weight are combined in a certain way to affect the model output. We proposed A-Ptr-Net model structure is shown in Fig. 2.

The end-to-end structure provides a self-driven learning procedure. The model attains the reward based on the negative of total route distance, and then it verifies the feasibility of a generated sequence. For multiple customer nodes, A-Ptr-Net model does not require an explicit distance matrix, and it can automatically adapt to a trajectory sequence. It is clear that only one feed-forward trajectory structure can update the route distance on account of the new data.

A. Encoder

We define the drone logistics delivery problem as the graph attention network S , and S consists of a set of nodes $N = \{0, 1, 2, \dots, n\}$. x_i is the matrix eigenvalue of each corresponding node. The total drone logistics delivery route is represented as $\pi = \{\pi_1, \pi_2, \pi_3, \dots, \pi_n\}$, where $\pi_t \in \{1, \dots, n\}$, $\pi_t \neq \pi_i$. The stochastic strategy $p(\pi | S)$ of logistics delivery routes π is expressed by θ as follow:

$$p(\pi | S) = \prod_{t=1}^n p_{\theta}(\pi_t | S, \pi_{1:t-1}) \quad (16)$$

Encoder generates the embedded map of all input nodes, which is used as the input node together with the masking and context layer. They generate the training sequence π at the decoder side.

1) *Embedded Inputs*: In order to simplify the drone logistics delivery problem, customer nodes r_i are standardized as $\hat{r}_i = \frac{r_i}{\zeta}$, $0 < r \leq \zeta$. The standardized demand function is between 0 and 1. For any delivery route R_k , the corresponding total time length is T_k , where $\sum_{i \in R_k} r_i \leq \zeta$.

The hidden layer of the model is represented by different biases and weights to distinguish the output vectors of depot and customer nodes in the network. W_0^x and b_0^x represent the weight and bias of the depot, respectively. W_0^x and b_0^x represent the weight and bias of the depot, respectively. Correspondingly, W^x and b^x represent the weight and bias of the customer node. The representation of initial embedded inputs is as Eq. 17:

$$h_i^{(0)} = \begin{cases} W_0^x x_i + b_0^x & i = 0 \\ W^x [x_i, \hat{r}_i, \varepsilon_i] + b^x & i = 1, \dots, n \end{cases} \quad (17)$$

To distinguish the output vectors of delivery centers and customer demand nodes in the network, the hidden layer of the model is represented by different weights and bias. The initial embedding representation of the A-Ptr-Net model lays the foundation for the decoding process, so that the decoder can search the optimal parameter weight.

2) *Capacity Constraints*: Drone capacity, customer delivery demand and service time windows constitute the computational constraints. We draw the concept of remaining variables into establishing the state transition equation of A-Ptr-Net model. When the service time is t , the remaining capacity, remaining demands and remaining time are expressed as ζ_t , $\hat{r}_{i,t}$ and $\varepsilon_{i,t}$. The initial constraint state of the three variables is expressed as $\zeta_t = 1$, $\hat{r}_{i,t} = \hat{r}_i$ and $\varepsilon_{i,t} = \varepsilon_i$, where $t = 1$.

The state transition equations are updated as follows:

$$\zeta_{t+1} = \begin{cases} \max(\zeta_t - \hat{r}_{\pi_t, t}, 0) & \pi_t \neq 0 \\ 1 & \pi_t = 0 \end{cases} \quad (18)$$

$$\hat{r}_{i,t+1} = \begin{cases} \max(\hat{r}_{i,t} - \zeta_t, 0) & \pi_t = i \\ \hat{r}_{i,t} & \pi_t \neq i \end{cases} \quad (19)$$

$$\varepsilon_{i,t+1} = \begin{cases} \max(\varepsilon_{i,t} - T_{\pi_t, t}, 0) & \pi_t = i \\ \varepsilon_{i,t} & \pi_t \neq i \end{cases} \quad (20)$$

where π_t is a set of customer node at the decode step t . $\pi_t = 0$ means that the drone initial carrying capacity is 1, on the contrary, the drone capacity decreases with the increase of delivery.

B. Decoder

The function of decoder is to transmit static and dynamic data stream. Through the RNN decoding process serially, the hidden layer codes for whole sequence are summed and averaged. When encoding, RNN encodes the hidden state of previous layer and the input characters of current layer, but there is no such input when decoding.

1) *Decoder Context*: The model decoder context at time t is based on the encoder and the output before time t , that is, the matrix output of the previous node set π_{t-1} . The formula is as follows:

$$h_{(c)}^{(N)} = \begin{cases} [h^{(N)}, h_{\pi_{t-1}}^{(N)}, \zeta_t] & t > 1 \\ [h^{(N)}, h_0^{(N)}, \zeta_t] & t = 1 \end{cases} \quad (21)$$

where $[\cdot, \cdot, \cdot]$ is the horizontal catenation operator. $h_{(c)}^{(N)}$ is a three-dimensional vector used to represent the embedding of context node (c), and the superscript (N) is aligned with the embedded node $h_i^{(N)}$. Next, the multi-head attention mechanism is used to calculate the next context embedding node $h_{(c)}^{(N+1)}$, and a single query $q_{(c)}$ of each context node $h_i^{(N)}$ is calculated to represent the output. The computational procedure is as Eq. 22 to Eq. 24:

$$q_{(c)} = W^Q h_{(c)} \quad (22)$$

$$k_i = W^K h_i + W_d^K \hat{r}_{i,t} \quad (23)$$

$$v_i = W^V h_i + W_d^V \hat{r}_{i,t} \quad (24)$$

In Eq. 22 to 24, W^Q , W^K and W^V represent the corresponding parameter matrix, and the projections addition of h_i and $\hat{r}_{i,t}$ is equivalent to the projection associated with a single matrix W^K .

2) *Masking*: In order to ensure network to input regular vectors, masking nodes $u_{(c)j}$ are set for the input of variable length nodes. Because the depot can be accessed multiple times, it cannot be accessed more than twice in the same trip π_t , where setting $u_{(c)j} = -\infty$, $t = 1$, $\pi_t = 0$. If the customer demand is 0, or the customer demand exceeds the drone remaining capacity, then $u_{(c)j} = -\infty$, $\hat{r}_{i,t} = 0$, $\hat{r}_{i,t} > \zeta_t$ is defined. Eq. 25 is the masking procedure:

$$u_{(c)j} = \begin{cases} \frac{q_{(c)}^T k_j}{\sqrt{d_k}} & \text{if } j \neq \pi_{t'} \quad \forall t' < t \\ -\infty & \text{otherwise} \end{cases} \quad (25)$$

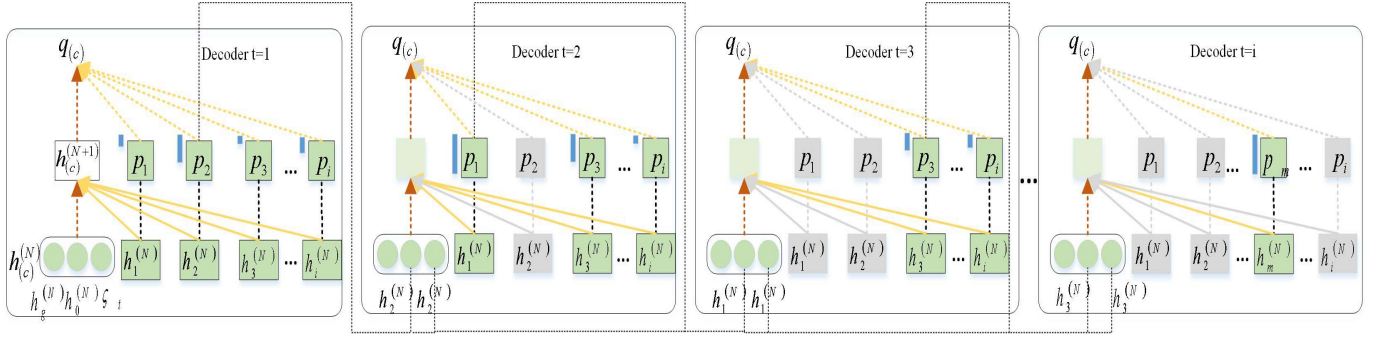


Fig. 3. Attention mechanism decoder for model.

C. Attention Mechanism

Encoder generates the embedded information of the input nodes with n attention layers in transformer architecture. Each attention layer contains two sub layers: multi-head attention layer and fully connected feed-forward layer. Fig. 3 shows the attention mechanism decoder for trajectory data. Attention mechanism can help decode in order. In step t , the decoder outputs the probability of selecting each node, according to the graph embedding and the output information generated at step t' . However, in order to improve the computational efficiency, not every node in the encoder interacts with other nodes. Only the node embedding information calculated is sent to the context node embedding of the existing information.

To calculate the probability trajectory output $p_\theta(\pi_t | \pi_{1:t-1})$ of Eq. 16, the attention head is added to the last layer of decoder. The layer compatibility $u_{(c)j}$ can be calculated based on Eq. 25, and its activation function $\tanh$ is expressed as follows:

$$u_{(c)j} = \begin{cases} C \cdot \tanh\left(\frac{q_{(c)}^T k_j}{\sqrt{d_k}}\right) & \text{if } j \neq \pi_{t'} \quad \forall t' < t \\ -\infty & \text{otherwise} \end{cases} \quad (26)$$

where $[-C, C]$ is the value range of the activation function. The above compatibility function is interpreted as the non-standardized logarithmic probability, and the probability of the final output vector p calculated by softmax function is Eq. 27.

$$p_i = p_\theta(\pi_t = i | S, \pi_{1:t-1}) = \frac{e^{u_{(c)i}}}{\sum_j e^{u_{(c)j}}} \quad (27)$$

V. EXPERIMENTAL RESULTS AND ANALYSIS

To evaluate the A-Ptr-Net deep reinforcement learning model for trajectory optimization, it will be applied in the randomly generated instances. The numerical training will compare the optimality gap between A-Ptr-Net model and other solutions. Moreover, it will discuss the impact of attention mechanism and optimizing nonlinear energy consumption.

A. Test Description

We test a specific capacitated version with one depot, one drone and multiple customer demand nodes for drone logistics delivery. Due to the limited payload of drone and the nonlinear

distribution of energy consumption function, drone needs to go back to the depot many times to complete the tasks of delivering parcels. Each customer demand node has a service time windows. This multitrip delivery with trajectory data exists tremendous optimizable parameters to fill the gap. The objective is to minimize the total route distance of drone completing the logistics delivery.

Delivery network nodes are randomly generated from the unit $[0, 1] \times [0, 1]$. For any a customer node, it follows a random distribution, and node 0 represents the depot. In addition, customer demand is also a random number belongs to $\{1, 2, \dots, 9\}$. The service time windows $[a_i, b_i]$ are randomly generated from the unit $[1, 9] \times [1, 9]$. If drone has a specific payload, the optimal drone delivery routing will be developed in a certain customer demand situation. By continuously assigning values to different instances, log of ratio and validation optimality gap can prove the combination optimization project. Therefore, the drone logistics delivery scheme generates a series of trajectory data. By training trajectory data, A-Ptr-Net model parameters are constantly fine-tuned.

B. Performance Comparison Between A-Ptr-Net Model and Other Solutions Based on Trajectory Data

For A-Ptr-Net model, two different decoders called greedy search and beam search are adopted. These searches train the feasible solutions from different aspects. Greedy search can obtain the highest possible customer node at the fastest speed, and it shortens the decoding time. Although the search output value is longer, beam search ensures the minimum route distance. Beam search generates the most probable routes by constructing beam width, which can expand the memory of candidate nodes.

To evaluate the proposed A-Ptr-Net model performance based on trajectory data, we compare the solutions obtained by some common exact and heuristic algorithms including mixed integer linear program (MILP), simulated annealing (SA), sweep coverage (SC), weighted targets sweep coverage (WTSC) and mobile-edge computing (MEC). We test the log of ratio under different drone capacity of each trip ζ and instance nodes. A-Ptr-Net model is decoded using greedy search and beam search, and others values have reported partly [50].

TABLE II
LOG OF RATIO WITH DIFFERENT SOLUTIONS

Capacity	Metrics	Method						
		PN-GS	PN-BS	MILP	SA	SC	WTSC	MEC
$\zeta=20$	Obj.	0.80	0.80	0.25	0.25	0.50	0.50	3.84
	Gap	3.98%	3.87 %	2.47 %	3.52 %	2.59 %	3.39 %	3.71 %
	Time	(0.24m)	(0.43m)	(0.97m)	(0.23m)	(0.62m)	(0.61m)	(0.81m)
$\zeta=30$	Obj.	0.80	0.80	1.00	1.00	0.70	0.70	3.85
	Gap	3.78%	3.21%	1.65%	2.67%	1.62%	2.41%	2.74%
	Time	(0.26m)	(0.43m)	(0.96m)	(0.25m)	(0.83m)	(0.65m)	(0.85m)
$\zeta=40$	Obj.	0.50	0.50	0.25	0.25	0.50	0.50	6.10
	Gap	2.84%	2.12%	1.23%	1.96%	1.28%	1.79%	1.35%
	Time	(0.31m)	(0.45m)	(1.05m)	(0.28m)	(0.90m)	(0.74m)	(0.89m)
$\zeta=50$	Obj.	0.50	0.50	1.00	1.00	0.70	0.70	6.14
	Gap	2.81%	2.07%	1.21%	1.94%	1.25%	1.69%	1.29%
	Time	(0.36m)	(0.49m)	(1.36m)	(0.28m)	(0.96m)	(0.96m)	(0.98m)

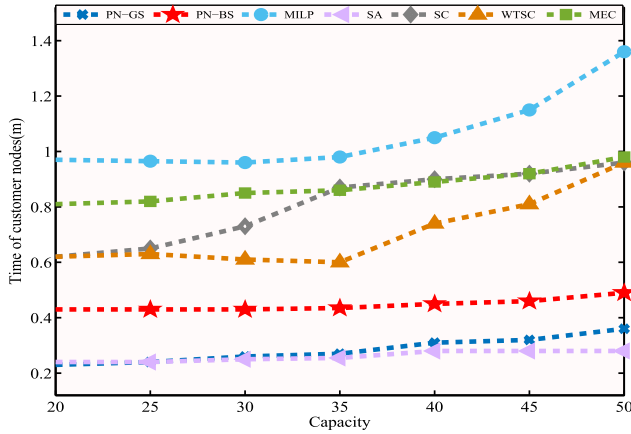


Fig. 4. Time of customer nodes with different methods based on trajectory data.

Table II is the results of drone with variable capacity from 20 to 50. It is of great significance to calculate the test object, optimality gap and run time. Under the guidance of different objectives, the metrics of each algorithm have quite changes. In order to compare the effect of decoding rules on A-Ptr-Net model, A-Ptr-Net with greedy search (PN-GS) and A-Ptr-Net with beam search (PN-BS) are both trained separately. On the whole, log of ratio presents a fixed variation characteristics. In order to compare the trend of different solutions more intuitively, we draw two figures about validation optimality gap and run time based on trajectory data.

Fig. 4 shows that time of customer nodes with different methods based on trajectory data are compared. Service time reflects the drone delivery agility. The shorter the service time of each node, the earlier it is possible to reach the next node. Drone can complete all the logistics delivery tasks with the fastest speed and the least energy consumption. The proposed A-Ptr-Net with greedy search and simulated

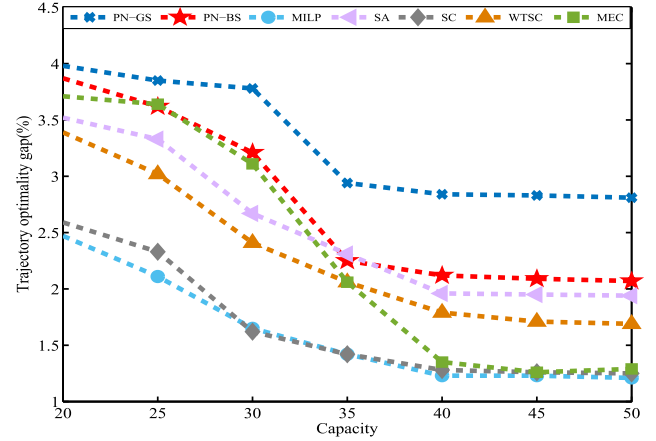


Fig. 5. Validation trajectory optimality gap with different methods.

annealing are at a relatively low time level. Although A-Ptr-Net with beam search takes more time than greedy search, it is still better than many other methods. For example, when the drone capacity reaches 50, time of customer nodes with PN-GS and PN-BS based on trajectory data is 0.36 and 0.49, respectively.

Validation trajectory optimality gap reflects the experimental test performance. As can be seen from Fig. 5, when the drone capacity is from 20 to 50, the optimality gap of all models is decreasing. This is because the increase of capacity brings more challenges, and drone may face more complex logistics environment. No matter how the capacity changes, the optimality gap of A-Ptr-Net model is always higher than other models. Deep reinforcement learning network structure has significant advantages in processing sequential decision. Especially, when drone capacity is 30, the optimality gap differences 0.17%. When drone capacity is 50, the optimality gap of A-Ptr-Net decoder is obvious.

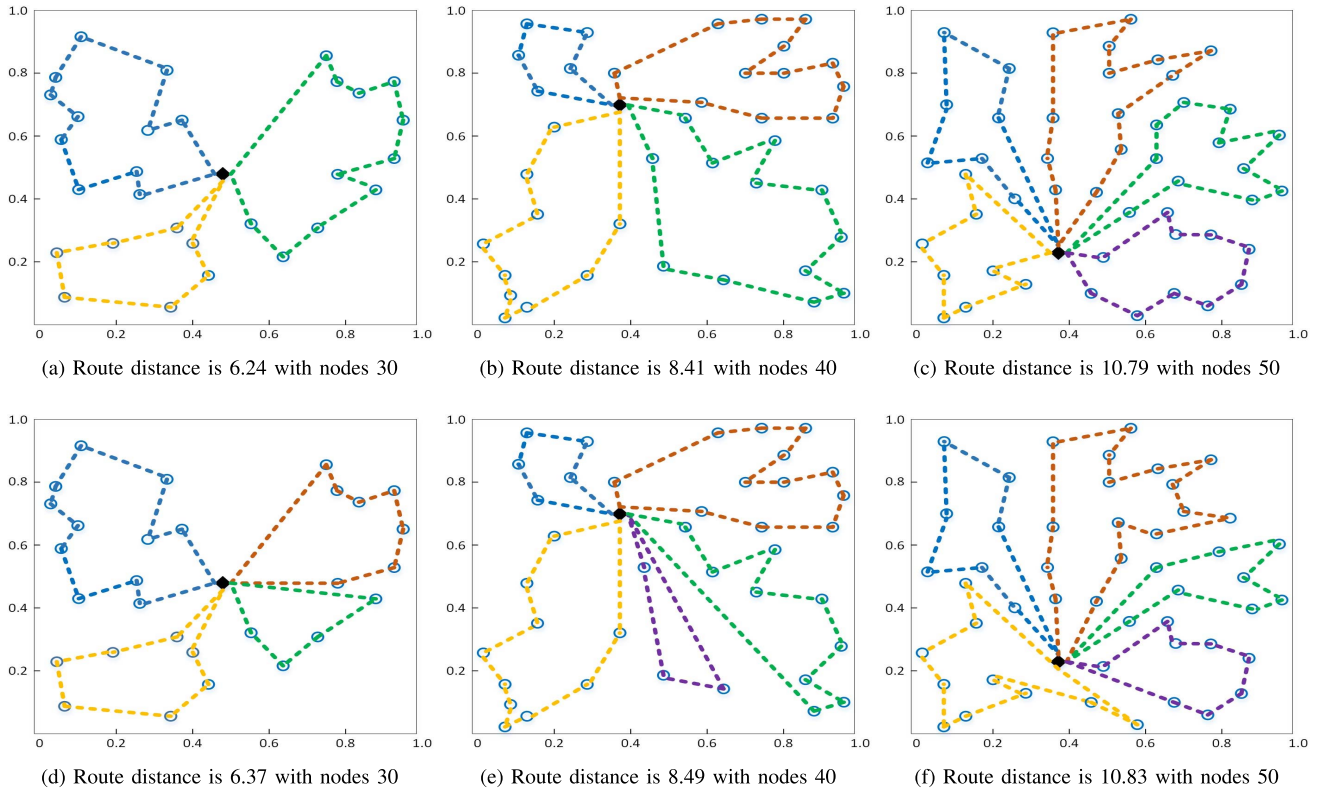


Fig. 6. Effect of attention mechanism for pointer network decoder ((a)-(c) is with attention mechanism, and (d)-(f) is without attention mechanism).

C. Effect of Attention Mechanism

To improve the search performance of interactive sequences, we add the attention mechanism to the pointer network decoder. Attention mechanism consists of multi-head attention layer and fully connected feed-forward layer. It allows the encoder to multiple output instead of using a single long vector of the corresponding step, so more information is retained. Attention mechanism helps the decoder to consider hidden state sequence of the whole encoder output, so that the graph embedding can store more information in all hidden state vectors. Important decode information is abstracted through hidden vectors, so network nodes can be sorted effectively.

We compare the route distance of drone completing the logistics delivery between attention mechanism and no attention mechanism. Network nodes and depot are randomly generated from the unit $[0, 1] \times [0, 1]$. When the drone capacity is the same, Fig. 6 shows the route distance with different decoding rules and the number of delivery network nodes. Attention mechanism shortens the delivery distance in overall feel, but effect of attention mechanism for A-Ptr-Net model decreases as the number of nodes increasing from 30 to 50. Drone go through logistics delivery according to the locations of customer node.

Specifically, attention mechanism can help reduce the drone route distance based on trajectory data. When the number of network nodes is 30, drone route distance is 6.24 with attention mechanism. However, in the absence of attention mechanism, drone needs to fly more 0.13 to complete the

same tasks. If we continue to increase the number of network nodes, drone delivery distance will increase accordingly. When there are enough nodes in the delivery network, the role of attention mechanism is not obvious. In Fig. 6(c) and Fig. 6(f), customer nodes add to 49, and attention mechanism can only reduce the distance by 0.04. In addition, network nodes are increased from 30 to 50, and the drone adds two trip. With the complexity of delivery network, drone needs to increase trip to meet the customer parcel demand.

D. Analysis of Optimizing Nonlinear Energy Consumption

We compare the solutions of linear and nonlinear energy consumption. Delivery network exists 10 supply and demand locations, and drone starts and finally returns to the depot. If the energy consumption function is evaluated linearly according to drone parameters, some energy gap will be lost. Energy consumption is related to drone payload, battery weight and own weight, which a complex nonlinear convex function. Table III shows that linear approximation has error with the actual situation although the calculation process can be simplified.

For the instance, the delivery trip of linear approximation is only once, and the optimal distance is 2.39. In fact, drone actually needs to deliver twice to meet all the customer needs. When drone completes the delivery of node 4, its energy consumption has reached the maximum, so it must return the depot for energy supply. Therefore, it is unscientific to make the linear approximation in this case. As can be seen from the energy gap, nonlinear energy consumption is also promising,

TABLE III
SOLUTIONS OF LINEAR AND NONLINEAR ENERGY
CONSUMPTION BASED ON TRAJECTORY DATA

Instance for nodes 10:
Depot location: [0.32,0.49]
Customer locations: [0.51,0.92], [0.63,0.27], [0.58,0.14], [0.82,0.75] and: [0.89,0.54], [0.16,0.39], [0.27,0.65], [0.33,0.74], [0.52,0.38]
Customer demands: [2,9,3,4,8,3,5,7,9]
Linear approximation:
PN-BS decoder trip: 0→9→3→2→5→4→1→8→7→6→0
Energy gap: 4.28
Optimal distance: 2.39
Nonlinear energy consumption:
PN-BS decoder trip: 0→9→3→2→5→4→0; 0→1→8→7→6→0
Energy gap: 4.63
Optimal distance: 3.24

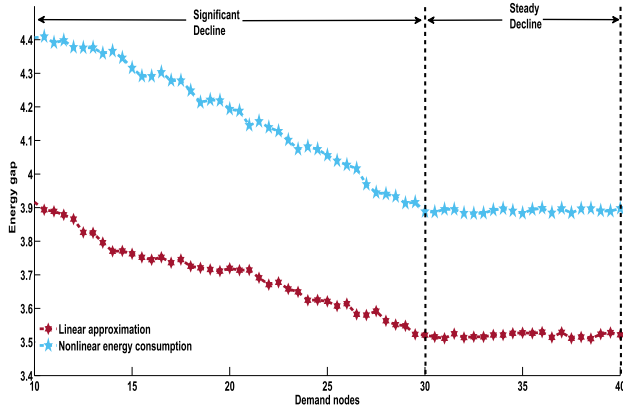


Fig. 7. Comparison of energy gap between linear approximation and nonlinear energy consumption.

even though the optimal distance is 3.24. Optimizing nonlinear energy consumption improve the performance of our proposed A-Ptr-Net model.

We make the comparison of energy gap between linear approximation and nonlinear energy consumption. Fig. 7 shows that the nonlinear energy consumption has obvious advantages with the different demand nodes. No matter how the demand nodes increases, the energy gap of linear approximation is always less than that of nonlinear energy consumption. When the number of demand nodes is less than 30, the energy gap of both drone energy computing methods has a significant decline. When the number of demand nodes exceeds 30, the function shows a steady decline trend. Therefore, keeping the nonlinear with the drone parameters is promising.

E. Discussions

Different from the traditional exact algorithm or a certain integrated heuristic algorithm, the A-Ptr-Net model with attention mechanism is more suitable for drone trajectory data. Moreover, the proposed reinforcement learning model

can derive the reward calculation and adapt to the new trajectory automatically, regardless of explicit distance matrix. Attention mechanism helps the decoder to consider hidden state sequence of the whole encoder output, so that the graph embedding can store more information in all hidden state vectors.

VI. CONCLUSION

In this paper, we propose a novel self-driven A-Ptr-Net reinforcement learning model to optimize drone trajectory data and find the optimal delivery path. The A-Ptr-Net model is constructed by a set of integral generated sequences, and the generated structure coupled with attention mechanism helps to adapt to the new trajectory structure automatically. Specifically, we develop the convex function constraints related to drone nonlinear energy consumption, customer demand and service time windows. After that, the proposed A-Ptr-Net model learns from trajectory optimization matrix. It makes use of recurrent neural network to map the trajectory data input to an embedding, and then it applies the graph embedding to store more information in the state vectors. Finally, attention mechanism is applied to adjust the trajectory decoder context. Experimental results show that the proposed model achieves superior performance than other algorithms for trajectory data. Regarding to future work, the logistics delivery of multiple drones should be analyzed to achieve more complex delivery network.

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